

ifm electronic



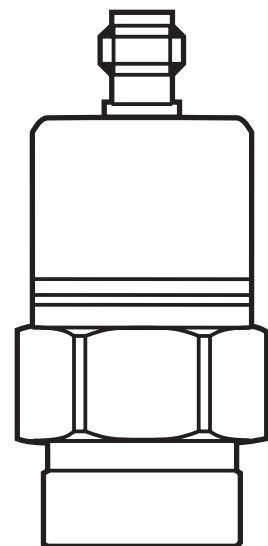
Installation Instructions
Electronic pressure sensor

efector500[®]

PA30xx / PA90xx

UK

704090 / 01 08 / 2010



Contents

1 Safety instructions	2
2 Function and features.....	3
2.1 Applications	3
3 Installation	4
4 Electrical connection.....	4
5 Scale drawing.....	5
6 Technical data.....	6

1 Safety instructions

- Please read the product description prior to installing the unit. Ensure that the product is suitable for your application without any restrictions.
- If the operating instructions or the technical data are not adhered to, personal injury and/or damage to property may occur.
- Please check for all applications that the product materials (see Technical data) are compatible with the media to be measured.
- Use in gases at pressures > 25 bar only after contacting the manufacturer ifm.

For units with cULus approval and the scope of validity cULus: The device shall be supplied from an isolating transformer having a secondary Listed fuse rated as noted in the following table.

Overcurrent protection		
Control-circuit wire size		Maximum protective device rating Ampere
AWG	(mm ²)	
26	(0.13)	1
24	(0.20)	2
22	(0.32)	3
20	(0.52)	5
18	(0.82)	7
16	(1.3)	10

The Sensor shall be connected only by using any R/C (CYJV2) cord, having suitable ratings.

2 Function and features

The pressure sensor detects the system pressure and converts it into an analog output signal.

- 4 ... 20 mA (PA30xx) / 0 ... 10 V (PA90xx)

2.1 Applications

- Type of pressure: relative pressure

Order no.	Measuring range		Permissible overload pressure		Bursting pressure	
	bar	PSI	bar	PSI	bar	PSI
PAxx20	0...400	0...5 800	600	8 700	1 000	14 500
PAxx21*	0...250	0...3 625	400	5 800	850	12 300
PAxx22	0...100	0...1 450	300	4 350	650	9 400
PAxx23	0...25	0...363	150	2 175	350	5 075
PAxx24	0...10	0...145	75	1 087	150	2 175
PAxx26	0..2.5	0...36.3	20	290	50	725
PAxx27	0...1	0...14.5	10	145	30	450
PAxx28	0...0.25	0...3.63	10	145	30	450
PAxx29	-1...0	-14.5...0	10	145	30	450
PAxx60	0...600	0...8 700	800	11 600	1 200	17 400

UK

MPa = bar ÷ 10 / kPa = bar × 100

*Values for PA3221:

Permissible overload pressure = 600 bar; bursting pressure = 1000 bar.



Static and dynamic overpressures exceeding the indicated overload pressure are to be avoided by taking appropriate measures.

The indicated bursting pressure must not be exceeded. Even if the bursting pressure is exceeded only for a short time, the unit can be destroyed.

NOTE: Risk of injury!

Use in gases at pressures > 25 bar only after contacting the manufacturer ifm..

High-pressure units (400 bar, 600 bar) are supplied with an integrated damping device to avoid any risk of injury in case of bursting when bursting pressure is exceeded.

When the damping device is removed it can become unusable.

If you have any questions, please contact ifm electronic's sales specialists.

3 Installation



Before mounting and removing the sensor, make sure that no pressure is applied to the system.

- ▶ Insert the unit in a G $\frac{1}{4}$ process connection.
- ▶ Tighten firmly.

4 Electrical connection



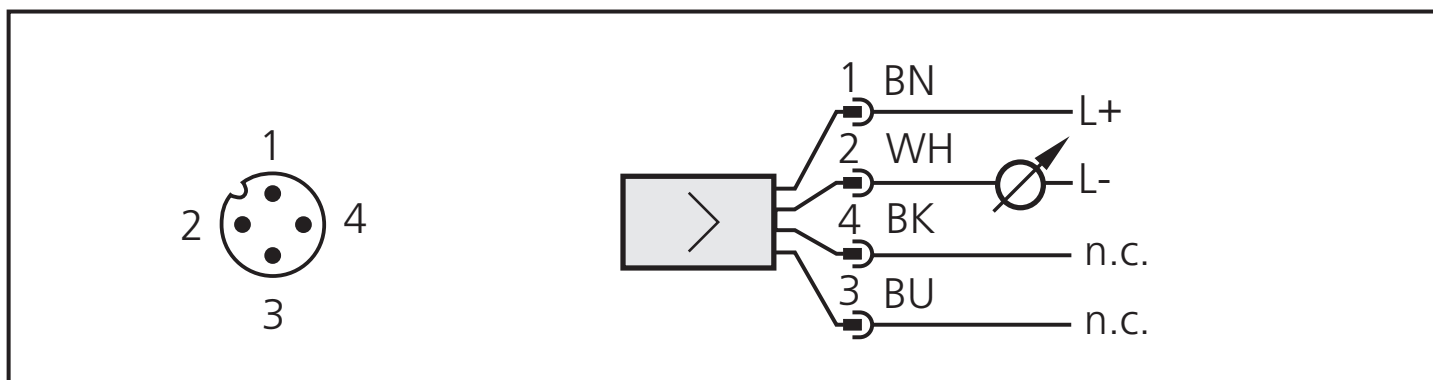
The unit must only be connected by an electrician.

The national and international regulations for the installation of electrical equipment must be observed.

Voltage supply to EN50178, SELV, PELV.

- ▶ Disconnect power.
- ▶ Connect the unit as follows:

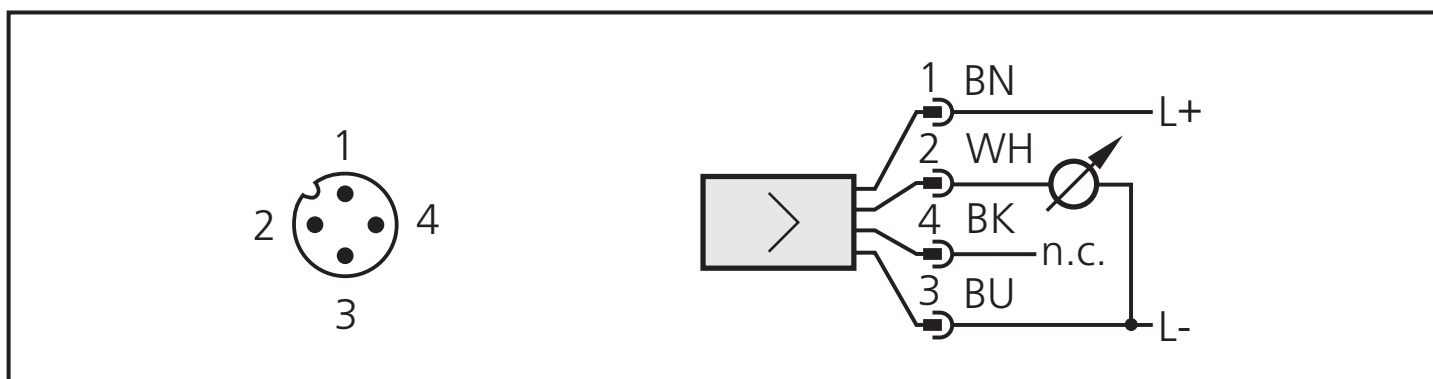
PA30xx (4 ... 20 mA analogue)



Core colours of ifm sockets:

1 = BN (brown), 2 = WH (white), 3 = BU (blue), 4 = BK (black),
n.c. = not connected.

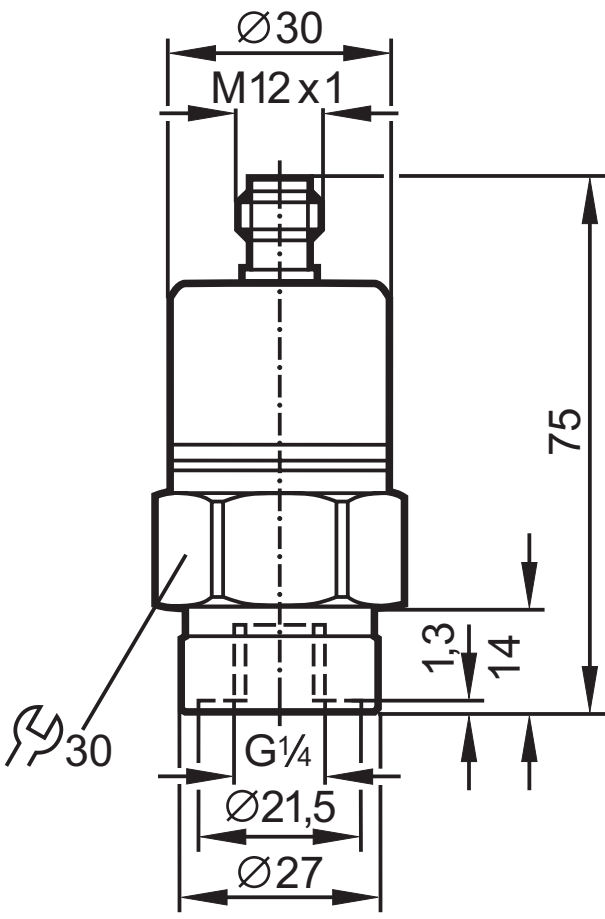
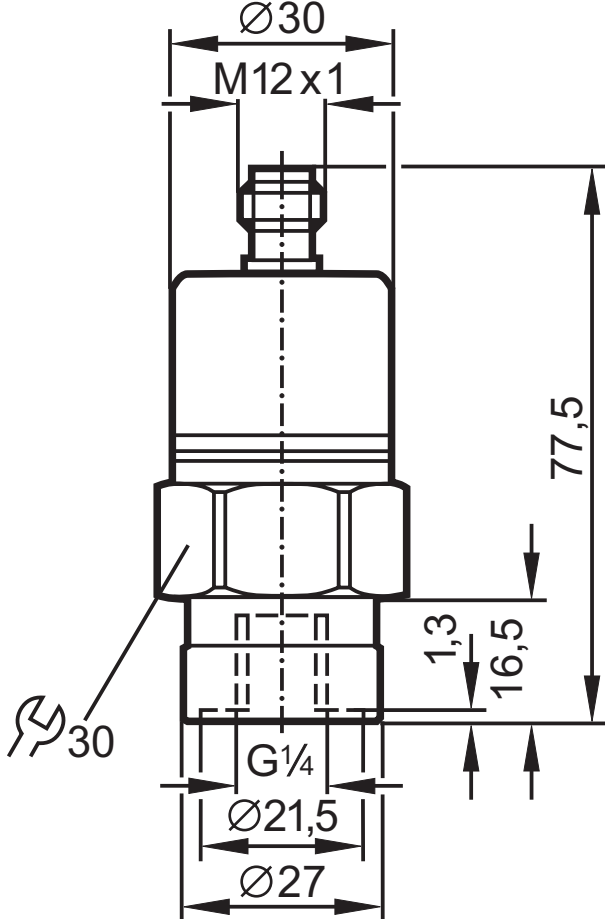
PA90xx (0 ... 10 V analogue)



Core colours of ifm sockets:

1 = BN (brown), 2 = WH (white), 3 = BU (blue), 4 = BK (black),
n.c. = not connected.

5 Scale drawing

PAxx21 ... PAxx29	PAxx20, PAxx60
	

Dimensions are in millimeters

UK

6 Technical data

PA30xx		
Operating voltage [V].....		9,6 ... 32 DC ¹⁾
Analogue output		4 ... 20 mA
Load [Ω].....		max. (UB - 9.6) x 50; 720 at UB = 24 V
PA90xx		
Operating voltage [V].....		16 ... 32 DC ¹⁾
Current consumption [mA].....		< 18
Analogue output		0 ... 10 V
Load [Ω].....		min. 2000
Step response time analogue output [ms].....		3
Characteristics deviation [% of the span]		< ± 0.25 (BFSL) / < ± 0.5 (LS)
Repeatability [% of the span].....		< 0.1
Long-term stability [% of value of measuring range / 6 months].....		< ± 0.05
Temperature coefficients (TEMPCO) in the compensated temperature range 0 ... 80°C (in % of the span/10K) ²⁾		
	PAxx20...PAxx27, PAxx60	PAxx28
greatest TEMPCO of the zero point	0.1	0.2
greatest TEMPCO of the span	0.2	0.3

Housing material.....	stainless steel (316S12); FPM (Viton); PA; EPDM/X (Santoprene)
Materials (wetted parts).....	stainless steel (303S22); ceramics; FPM (Viton)
Operating temperature [°C]	-25 ... +80
Medium temperature [°C]	-25 ... +90 ²⁾
Storage temperature [°C].....	-40 ... +100
Protection PAxxx0 ... PAxxx2, PAxx60	IP 68 / IP 69K
Protection PAxxx3 ... PAxxx9	IP 65
Protection class	III
Insulation resistance [MΩ]	> 100 (500 V DC)
Shock resistance [g]	50 (DIN / IEC 68-2-27, 11 ms)
Vibration resistance [g]	20 (DIN / IEC 68-2-6, 10 - 2000 Hz)
EMC	
EN 61000-4-2 ESD:.....	4 kV CD / 8 kV AD
EN 61000-4-3 HF radiated:	30 V/m
EN 61000-4-4 Burst:.....	2 kV
EN 61000-4-6 HF conducted:.....	10 V
radiation of interference	according to the automotive directive 2004/104/EC / CISPR25
noise immunity.....	according to the automotive directive 2004/104/EC / ISO 11452-2
HF radiated.....	100 V/m
pulse resistance.....	according to ISO7637-2 / severity level 3

¹⁾ to EN50178, SELV, PELV

²⁾ -40...90°C upon request

BFSL = Best Fit Straight Line / LS = Limit Value Setting

More information at www.ifm.com

UK